

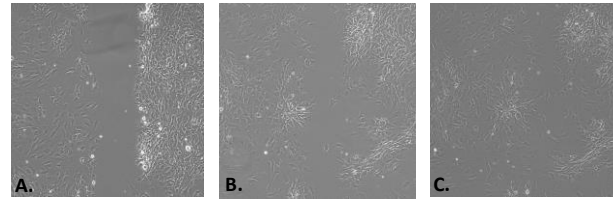
Cannabinoid Agonist WIN-55, 212-2 Modulation of Bone Derived Mesenchymal Stem Cells

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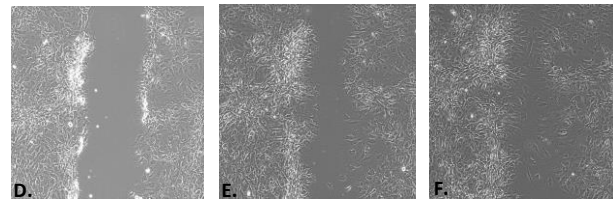
Background

Mesenchymal stem cells (MSC's) are multipotent stromal cells with the ability to differentiate into several types of cells including bone, adipose tissue, cartilage, muscle cells and neuron-like cells. MSC's may be a viable option for transplantation into a host due to unique properties; they are immunosuppressive and are more likely to be accepted by the host than other types of stem cells. However, survival and functional integration of MSC's still poses a difficult problem for effective transplantation into a host. Unique methods are needed to improve MSC application in vivo. Cannabinoid receptors are present in MSC's and may influence cell proliferation, migration, differentiation and functional integration. Anti-inflammatory and neurotrophic mechanisms of cannabinoids may modulate MSC's leading to increased survival and efficacy during transplantation. The synthetic full CB₁ receptor agonist WIN-55 was administered to determine if cannabinoids influence cell proliferation, migration and CB₁ quantity in a dose dependent manner.

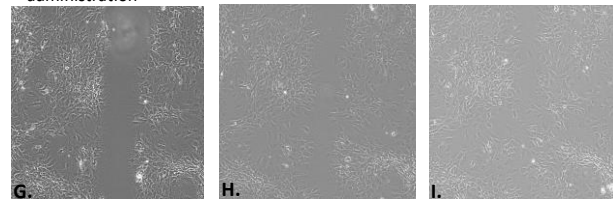
MSC Migration



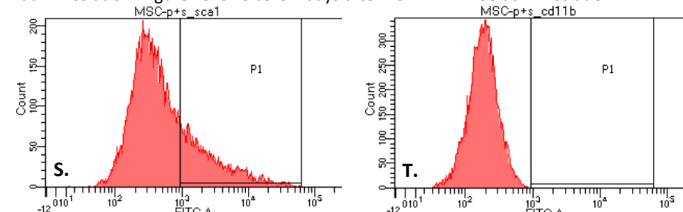
DMSO (control) group: Figure A shows cells before drug administration once cells were confluent (72 hours). Figure B shows cells 3 days after DMSO administration. Figure C shows cells 7 days after DMSO administration



1mM WIN-55 group: Figure D shows cells before drug administration once cells were confluent (72 hours). Figure E shows cells 3 days after 1mM WIN-55 administration. Figure F shows cells 7 days after 1mM WIN-55 administration



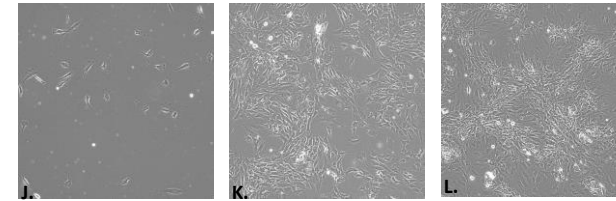
10mM WIN-55 group: Figure G shows cells before drug administration once cells were confluent (72 hours). Figure H shows cells 3 days after 10mM WIN-55 administration. Figure I shows cells 7 days after 10mM WIN-55 administration



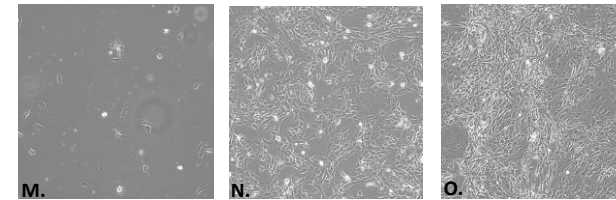
Fluorescence Activated Cell Sorting (FACS) was used to identify if cell surface markers on MSC's were present in this experiment. Figure S shows that sca1 was present (a marker of MSC's) and figure T shows that CD11b was not present (not a marker for MSC's). Along with morphology from imaging, this indicates that MSC's were used.

Figure U shows the group cell quantity means across each dosage group, at 3 days (blue/bottom) and 7 days (green/top) WIN-55 injection. This graph indicates that the 1mM WIN-55 group showed significantly more cell proliferation than the control group (DMSO only) and the 10mM WIN-55 group.

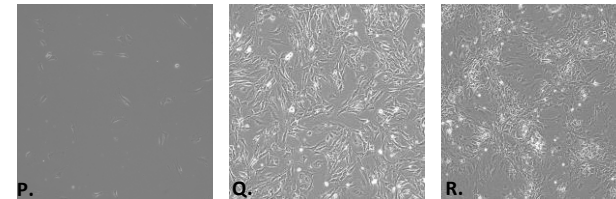
MSC Proliferation



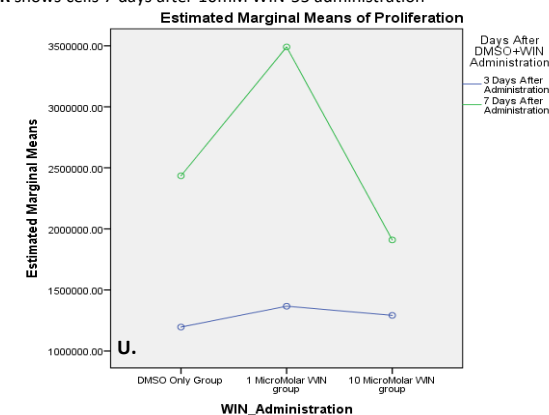
DMSO (control) group: Figure J shows cells before drug administration 24 hours post-passage. Figure K shows cells 3 days after DMSO administration. Figure L shows cells 7 days after DMSO administration



1mM WIN-55 group: Figure M shows cells before drug administration 24 hours post-passage. Figure N shows cells 3 days after 1mM WIN-55 administration. Figure O shows cells 7 days after 1mM WIN-55 administration



10mM WIN-55 group: Figure P shows cells before drug administration 24 hours post-passage. Figure Q shows cells 3 days after 10mM WIN-55 administration. Figure R shows cells 7 days after 10mM WIN-55 administration



- Passaged 50k cells into 25 cubic centimeter flasks. After 24 hours and adherence of cells to plastic, DMSO (25μl or DMSO+WIN-55 at either 1mM or 10mM was administered. Groups were also split into 3 and 7 post-drug administration periods.
- Preliminary results suggest that the application of the cannabinoid agonist WIN-55 increases cell proliferation in a dose dependent manner
- This study will determine if WIN-55 increases the rate of cell migration using scratch assays
- This study will evaluate if WIN-55 administration up-regulates the CB₁ receptor using western blot analysis
- Immunofluorescence of the CB₁ receptor will be evaluated using stereological techniques
- Results of this study will determine if in vivo applications will be pursued in future experiments.